STAT 223: Applied Analytics
Spring 2025

Instructor Information

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Office Hours: Before Class or by Appointment

Class Information

Dates: March 25 – June 2
Time: Period 5: 1:20 - 2:30 p.m.
Classroom: Science + Math Center - A203

Course Description

This course teaches students several of the main methodologies that are used in Applied Analytics, and gives them experience in doing analyses of large real (or realistic) data sets. Specifically, students will learn to:

1. predict values of continuous response variables using ordinary regression;
2. form models for time series data that enable forecasting of future values of the series;
3. use historical data on the dependence of a binary response (success or failure) on several possible predictors to classify new data as probable success or failures.

For each of these types of problems, they will learn how to carry out the analysis using appropriate technological tools, such as R or Python. Ethical issues in the use of analytics are also addressed.

Course Objectives

By the end of this course, you should be able to

- Formulate an empirical model for a set of data.
- Employ a variety of technologies for data exploration, analysis, and prediction.
- Ask appropriate questions relevant to a problem and a set of data
- Understand the ethical implications and limitations of data analysis (including issues relating to data security and privacy, possible biases in how data is obtained and analyzed, and ethical dangers of basing policy on the results of analysis).
- Communicate the results of an analysis effectively in writing.

Textbook, Calculators, & Software

Required Textbook: zyBooks: *STAT 223: Applied Analytics*

We will be using an online book from OpenStax for this course. It is a free OER. https://openstax.org/details/books/principles-data-science
We will cover much of the entire book. Topics to be covered include data visualization (descriptive analytics), regression, time series, supervised learning, and decision tree learning (predictive analytics), Monte Carlo methods and unsupervised learning (prescriptive analytics), as well as basic techniques in data analysis. See the tentative schedule for more details.

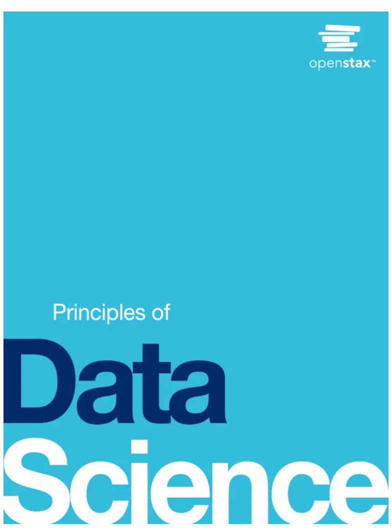
The OpenStax book covers the material in a high-level, light way. It should serve well as a guide through the course. We may go deeper into the material in class, or use supplementary material from other OERs.

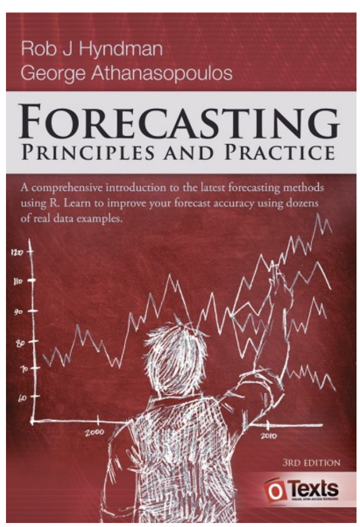
Supplementary Textbook: Hyndman, R.J., & Athanasopoulos, G. (2021) Forecasting: Principles and Practice, 2nd edition, OTexts: Melbourne, Australia. OTexts.com/fpp2.

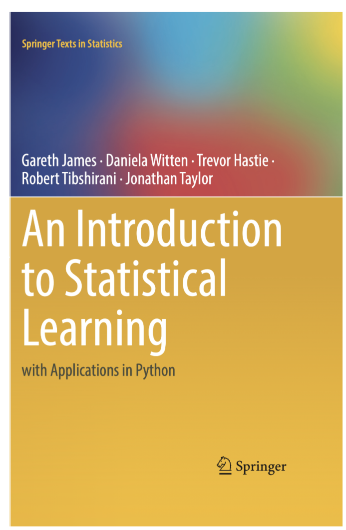
This is a supplement to support us in time series concepts, which the OpenStax book is light on. It is fully free, and we won't cover everything, but you should save the link even after it, as it's a great resource. The book is written in R.

Supplementary Textbook: James, G., Witten, D., Hastie, T., Tibshirani, R.,, Taylor, J. (2023). An Introduction to Statistical Learning with Applications in Python. Cham: Springer. ISBN: 978-3-031-38746-3 https://www.statlearning.com/

This book is a great book, it's often used as a text in statistical learning courses (I have a copy of the R version from a grad school course), but it is a bit too dense for our purposes. However, if there is anything you want to learn more about, it's a great place to start. It also has a ton of code and examples, and I may use some of these for our assignments. The PDF and supplementary material is completely free, as in speech and as in beer.







Calculator: You may need a calculator to do the computations that will arise. No specific calculator is required; I recommend the TI-30XS MultiView (https://a.co/d/0QHmuee). It's cheap and can do anything you need to in this course. You don't need a graphing calculator this course, but if you have a TI-84 already, it's more than capable. Note that if you use your phone as a calculator, you will need to find something for the exam, phones are not allowed during exams.

Software: We will be using R (and RStudio) as a review from STAT 200 at the start of this course, and mainly Python for this course, switching back to R for time series, as I feel the time series tools in R are easier to use/understand. I will use Google Colab to share Python notebook files, but you can use any program that puts together Python notebooks, if you are familiar with other platforms. You will need a computer for this class, and there will be a times when we look at code in class, so bringing a laptop to class if you are able would be advisable.

Class Attendance and Participation

It is essential to your success in this course that you attend and participate in each lecture. I will not take attendance directly, however there is an element of your grade that will include in-class elements. In my experience as a student, a teaching assistant and as a lecturer, the number one most common problem students have in courses is falling behind in the course and not being able to catch back up. If you are forced to miss a day, it is your responsibility to catch yourself back up, whether through getting notes from a classmate or coming to me to go over the material.

In class, I ask that you respect my time, your time and your classmate's time. In class, please avoid anything that would be too distracting your classmate: no cell phones, music, loud foods, etc. I personally recommend taking notes with pen and paper, but if you prefer using a laptop/tablet, please stay focused on the lecture and don't go surfing the web. (Do kids still say that?)

I aim to create an open and welcoming environment for learning, so if you have questions or don't understand something I've said, please speak up. If you are confused about something, it's almost certain that you're not the only one, and I'm happy to clarify. If you notice a mistake in my math, let me know, I'm only human and I promise I've made more stupid math mistakes than you have! And if you're curious or want to know more about a subject, speak up, I got my PhD in Mathematics & Statistics because I love this stuff!

Email Policy

If you have any questions outside of class, you can stop by my office, but you can also email me. I will attempt to respond to all emails within 24 hours. If you send an email late at night, I may not respond until the next day, and if you email on the weekends, I may not respond until the following Monday. If I fail to respond after 24 hours during the week, please resend the email, as I may have just not noticed it. I recommend that you email me using your Knox College email account, and keep note that there may be topics I may not be able to discuss over email or feel comfortable discussing over email, in which case I will make a time for us to meet in person to speak.

Academic Integrity & Honor Code

Remember that all work you submit must be in accordance with the Knox College Honor Code (https://www.knox.edu/offices/academic-affairs/honor-code-and-procedures/). Should I find evidence of a violation of academic integrity, I will sanction you.

Note two passages from the Honor Code in particular:

> *Students are expected to leave books and notes elsewhere during an examination and to take examinations in public places such as classrooms or open lounge areas nearby in the buildings. Washrooms, storage areas, and maintenance areas in the basement and elsewhere are not public places. Neither are carrels or other closed areas . . .*
>
> *Faculty may require that students not bring books, notes or electronic devices into the building in which an examination is administered. Alternatively, they may permit students to bring these resources into the building but may require that these books, notes and electronic devices be left in a central area or the classroom throughout the duration of the examination or other academic obligation.*

In accordance with these policies, you should not have a cell phone on you when taking exams or quizzes. I will not allow computers or other electronic devices, notes, textbooks, or any other help on exams.

Homework can be done cooperatively, but any submissions must be in your own words.

I encourage all of you to form weekly study groups to discuss the material and any problems you are having. Keeping up with the material will help you from falling behind through out the term, especially with how fast terms move here, and having the help holding each other accountable and helping each other is a great way to help with that.

AIs such as ChatGPT are expressly forbidden for homework assignments, both because they will not help you learn (the only way to learn mathematics is through doing mathematics), and because it doesn't even answer correctly most of the time, so don't waste your time. You may use ChatGPT as a resource alongside Google for coding assistance, but you also need to understand what your code is doing and how you would recreate it, and you will be tested on your knowledge.

Office of Disability Support Services

The Office of Disability Support Services website can be found at https://www.knox.edu/offices/disability-support-services.

Knox College abides by Section 504 of the Rehabilitation Act of 1973 (https://www.epa.gov/external-civil-rights/section-504-rehabilitation-act-1973), which stipulates:

> *No otherwise qualified individual with a disability in the United States, as defined in section 7(20) shall, solely by reason of her or his disability, be excluded from the participation in, be denied the benefits of, or be subjected to discrimination under any program or activity receiving Federal financial assistance or under any program or activity conducted by any Executive agency or by the United States Postal Service.*

Disabilities covered by law include, but are not limited to, learning disabilities, psychological disabilities, health impairments, hearing, and sight or mobility impairments.

If you have a disability that may have some impact on your work in class and for which you may require accommodations, please contact Stephanie Grimes in the Office of Disability Support Services (office: SMC E-115; email: sgrimes@knox.edu) so that such accommodations may be arranged.

Note that the accommodations are not retroactive; I do not make adjustments until ODSS contacts me to let me know what I can do. Also note that accommodations must be renewed each term.

Tutoring

The Center for Teaching and Learning (CTL) offers many resources for students in need of help. Asking for help is no reason to be embarrassed; seeking help in a big part of the learning process, and a sign that you are engaging with the material and your coursework. You are not alone at Knox College and you should utilize all the tools at your disposal to succeed.

Writing Tutors: If you want help on a paper, no matter the course, you can find writing tutors in the Writer's Workshop, which is at the Center for Teaching and Learning (466 S. West St., across from Post Hall), Monday through Friday, from 10:30 to 4:00pm, and again in Red Room on Tuesday, Wednesday, and Thursday nights from 7:00 to 9:00pm. You can also schedule appointments to meet with a writing tutor either online or in person (should you both so desire) by visiting the Center for Teaching and Learning website.

Subject Tutors: The Red Room study tables are great resources for students wanting help with virtually any subject— including STAT 200. There are two Red Room study areas: "Red Room" and "Red Room SMC." Red Room SMC is intended to support most classes taught in SMC and is located in the Learning Commons in SMC (under the whale), while Red Room is for most other courses and is located in the Red Room on the second floor of Seymour Library. Both study areas are free and open on a walk-in basis from 7:00pm to 9:00pm, on Tuesday, Wednesday, and Thursday nights.

TimelyCare

If you're unfamiliar with mental health support, taking care of your mental well-being can be stressful, or even scary. Knox is partnered with TimelyCare to provide a virtual health care solution that reflects Knox's values and student population. It is a student-centered platform that offers on-demand access to mental healthcare and a diverse, culturally competent provider network. TimelyCare complements Knox's on-campus services with 24/7 on-demand virtual mental health care, health coaching, peer to peer support and a plethora of coping strategies ranging from homesickness to depression to study skills. There are 12 free sessions available from September to August annually and can be accessed anywhere on or off campus (even internationally!). To access TimelyCare, go to TimelyCare.com or use the QR code below, register with your Knox email and follow prompts! If you have any questions, please reach out to Knox Health and Wellness at counseling@knox.edu or (309) 341-7492. If there is an emergency, please call 911 or contact Campus Safety at (309) 341-7979.

Title IX

Knox College values a culture of respect. If you have encountered any form of discrimination or harassment, you are encouraged to report it at https://www.knox.edu/about-knox/our-values/culture-of-respect/report-it.

As a faculty member, I am required to disclose to the Director of Title IX & Civil Rights Compliance any information I receive about discrimination, harassment, sexual misconduct and other forms of interpersonal violence. This includes information shared in class assignments, discussions, emails, and/or shared conversations outside of class. If a disclosure is received, she will reach out to you. You are not required to respond to this outreach.

If you wish to speak with a confidential resource (one not required to report to Title IX), please contact:

- Knox College Health Services (https://www.knox.edu/offices/health-and-counseling-services/health-services)
- Knox College Counseling Services (https://www.knox.edu/offices/health-and-counseling-services/counseling-services)
- Knox College Director of Spiritual Life (https://www.knox.edu/offices/spiritual-life/meet-the-staff)
- TimelyCare (https://app.timelycare.com/auth/login)

Pregnancy and/or Pregnancy Related Conditions

Faculty members are required to inform a student who is pregnant or experiencing a pregnancy related condition (including termination) that the Director of Title IX & Civil Rights Compliance is available to provide support by helping you to understand your rights/options and provide supportive measures. You can contact the Director of Title IX & Civil Rights Compliance at cultureofrespect@knox.edu.

Grading Information

Exams

We will have two midterm exams in this course, tentatively in the 5th and the 8th week of the course. The exact date will be confirmed as we approach the date. Each test will account for about **15% of your grade**, for a total of about **30% of your grade**.

At the end of the course, we will have a cumulative final exam during the time scheduled by the registrar's office. It will count for about **20% of your grade**.

Exams will be given in the classroom that we do lectures in. I have spoken with Dean Crawford, and I will be expecting everyone to take the exam in the room, unless you have an accomodation and prior approval. A calculator is all you can use on the exam; no computers, no phones, no notes, not talking to other classmates.

Programming Labs

There will be regular programming assignments as well as some additional pencil-paper work. They will be assigned as an PDF document, the format will be decided later, and will be posted to the Classroom page. The lengths of these assignments may also vary, and some may indeed rise to the level of "projects". These assignments will account for a total of about **30% of your grade**.

Attendance & In-Class Work

At various times during the terms, I will assign various types of in-class work, whether that be a small programming assignment, working through a problem or a small quiz. These assignments will account for a total of about **10% of your grade**.

Participation Activities

I plan to assign small assignments to accompany each lecture to help understand the material. Late homework will not be accepted unless there are extenuating circumstances – and you must inform me of these circumstances before the deadline. These activities will account for a total of about **10% of your grade**.

Grading Breakdown

Thus the grades will break down approximately as follows:

Component	Points
Midterm Exam #1	15%
Midterm Exam #2	15%
Final Exam	20%
Programming Assignments	30%
Attendance & In-Class Work	10%
Participation Activities	10%

Grade Scale

Final grades will be assigned according to the following point totals, though I reserve the right to alter the ranges as need be:

Grade	Range
A	>93%
A-	90% – 93%
B+	87% – 90%
B	83% – 87%
B-	80% – 83%
C+	77% – 80%
C	73% – 77%
C-	70% – 73%
D+	67% – 70%
D	63% – 67%
D-	60% – 63%
F	<60%